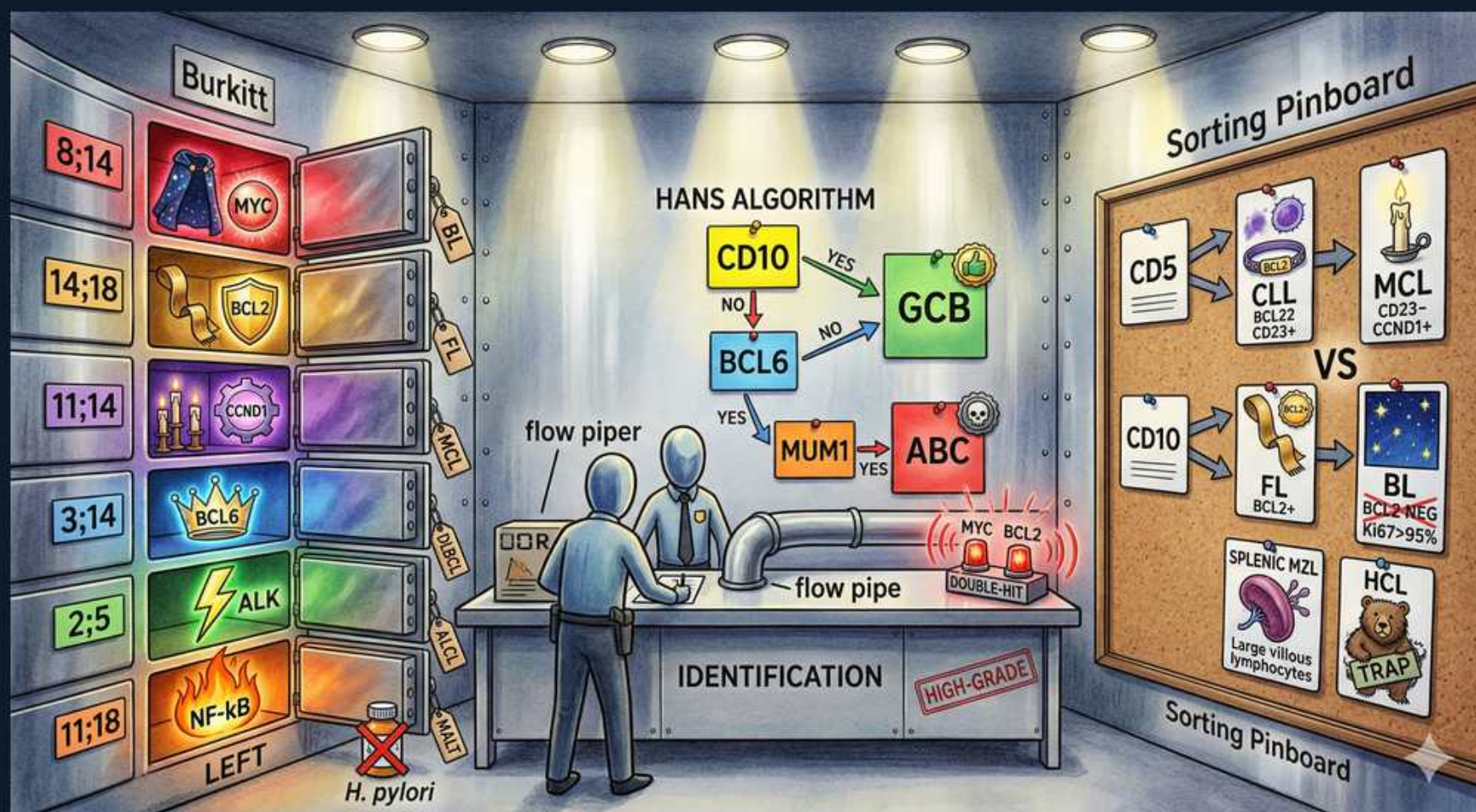


Lymphoma Translocations & Immunophenotype Sorting



1. t(8;14) MYC — Burkitt

WHAT YOU SEE

8;14 vault, MYC

WHAT IT ENCODES

t(8;14) juxtaposes MYC to the IgH promoter → relentless proliferation. Defines Burkitt lymphoma (also variants t(2;8), t(8;22)). Ki-67 ~100%, starry-sky histology.

BOARD TRAP

MYC alone = Burkitt; MYC plus BCL2 and/or BCL6 = double/triple-hit high-grade B-cell lymphoma, a distinct worse-prognosis entity.

2. t(14;18) BCL2 — Follicular

WHAT YOU SEE

14;18 vault, BCL2

WHAT IT ENCODES

t(14;18) places BCL2 under IgH control → anti-apoptosis, the hallmark of follicular lymphoma. Indolent, incurable, prone to transformation. Also seen in many GCB-DLBCL.

BOARD TRAP

BCL2 blocks apoptosis (survival), not proliferation — that's why FL is indolent, unlike proliferation-driven Burkitt.

3. t(11;14) CCND1 — Mantle cell

WHAT YOU SEE

11;14 vault, CCND1

WHAT IT ENCODES

t(11;14) drives cyclin D1 (CCND1) overexpression — cell-cycle progression. Defines mantle cell lymphoma; immunophenotype CD5+, CD23-, cyclin D1+, SOX11+.

BOARD TRAP

CD5+ but CD23- and cyclin D1+ = mantle cell; CD5+ CD23+ = CLL/SLL. The CD23 and cyclin D1 distinguish them.

4. t(3;14) BCL6 — GCB / DLBCL

WHAT YOU SEE

3;14 vault, BCL6

WHAT IT ENCODES

BCL6 rearrangements (often t(3;14)) are seen in DLBCL and follicular lymphoma; BCL6 is a germinal-center master regulator. Part of double/triple-hit definitions.

BOARD TRAP

BCL6 is a marker of germinal-center origin — its presence supports GCB phenotype on Hans algorithm.

5. t(2;5) ALK — ALCL

WHAT YOU SEE

2;5 vault, ALK

WHAT IT ENCODES

t(2;5) NPM-ALK defines ALK+ anaplastic large cell lymphoma (T-cell, CD30+). ALK+ has better prognosis than ALK-. Hallmark cells with horseshoe nuclei.

BOARD TRAP

ALK+ ALCL is the favorable T-cell lymphoma; ALK-negative ALCL behaves worse and overlaps with other PTCL.

6. t(11;18) NF-κB — MALT

WHAT YOU SEE

11;18 vault, NF-κB

WHAT IT ENCODES

t(11;18) (API2-MALT1) activates NF-κB in gastric MALT lymphoma and predicts FAILURE of H. pylori eradication therapy → needs radiation/systemic treatment.

BOARD TRAP

t(11;18)+ gastric MALT will not regress with antibiotics — don't keep treating H. pylori alone.

7. Hans algorithm: CD10

WHAT YOU SEE

CD10 yellow box, YES→GCB

WHAT IT ENCODES

Hans algorithm classifies DLBCL cell-of-origin by IHC. CD10+ → germinal center B-cell (GCB). GCB generally has better outcomes with R-CHOP than ABC.

BOARD TRAP

CD10 positivity routes straight to GCB — you don't even need BCL6/MUM1 if CD10 is positive.

8. Hans: BCL6 / MUM1 branch

WHAT YOU SEE

BCL6 blue, MUM1

WHAT IT ENCODES

If CD10-negative: BCL6- → ABC (non-GCB). If BCL6+ then MUM1- → GCB, MUM1+ → ABC. ABC subtype is more aggressive, driven by chronic BCR/NF-κB signaling.

BOARD TRAP

ABC (non-GCB) DLBCL does worse and is the target of novel BTK/NF-κB-directed agents; don't equate it with GCB.

9. Double-hit / high-grade

WHAT YOU SEE

MYC + BCL2 DOUBLE-HIT, HIGH-GRADE

WHAT IT ENCODES

High-grade B-cell lymphoma with MYC + BCL2 and/or BCL6 rearrangement — aggressive, CNS-prone, poor R-CHOP response; treated with dose-intensive regimens (e.g., DA-EPOCH-R).

BOARD TRAP

Double-hit is defined by translocations on FISH, not just protein 'double-expressor' by IHC — they're prognostically different.

10. CD5+ split: CLL vs MCL

WHAT YOU SEE

CD5 → CLL vs MCL

WHAT IT ENCODES

Both CLL and mantle cell are CD5+. CLL: CD23+, cyclin D1-, dim CD20. MCL: CD23-, cyclin D1+/SOX11+, t(11;14). This single fork is heavily board-tested.

BOARD TRAP

Default CD5+ to CLL is a trap — check CD23 and cyclin D1 to avoid missing aggressive MCL.

11. CD10+ split: FL vs Burkitt

WHAT YOU SEE

CD10 → FL vs BL, Ki67 95%

WHAT IT ENCODES

CD10 marks germinal-center origin (FL and Burkitt). FL: BCL2+, low Ki-67, t(14;18). Burkitt: BCL2-, Ki-67 ~95-100%, MYC t(8;14), starry sky.

BOARD TRAP

BCL2 positivity argues AGAINST Burkitt; a CD10+ tumor with Ki-67 near 100% and BCL2-negative is Burkitt.

12. Splenic MZL / villous lymphocytes

WHAT YOU SEE

SPLENIC MZL, villous lymphocytes

WHAT IT ENCODES

Splenic marginal zone lymphoma — massive splenomegaly, villous lymphocytes on smear, often HCV-associated, cytopenias from hypersplenism. Indolent; splenectomy or rituximab.

BOARD TRAP

Villous lymphocytes + splenomegaly + HCV → splenic MZL; treat HCV first if present.

13. Hairy cell leukemia (TRAP)

WHAT YOU SEE

HCL, TRAP

WHAT IT ENCODES

Hairy cell leukemia — pancytopenia, massive splenomegaly, dry tap, TRAP+/BRAF V600E, CD11c/CD25/CD103+. Treated with cladribine/pentostatin.

BOARD TRAP

Dry marrow tap + monocytopenia + splenomegaly = hairy cell; BRAF V600E and CD103 confirm it.

14. Flow cytometry identification

WHAT YOU SEE

Flow piper / flow pipe

WHAT IT ENCODES

Flow cytometry immunophenotyping is the central tool sorting B vs T, CD5/CD10/CD23/CD20 markers — driving every fork above. Combined with FISH for translocations.

BOARD TRAP

Morphology alone can't classify lymphoma — flow immunophenotype + FISH cytogenetics are required for diagnosis.